Maintenance Schedule

Frac Pack
EPA/CARB
TF 12V4000A1
TF 12V4000A2
Application Group 4D

MSE50215/01E

Power. Passion. Partnership.

System designation	Engine model	kW/cyl.	Application group
TF12V4000A1	12V4000S83	139,8 kW/cyl.	4D, Frac-operation (continuous operation with low load)
TF12V4000A2	12V4000S83L	155,4 kW/cyl.	4D, Frac-operation (continuous operation with low load)

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

The maintenance schedule is applicable to products certified in accordance with U.S. emissions regulations and operated in the area of applicability of these regulations.

Information about the latest maintenance schedule is available at: http://www.mtu-online.com.

The following instructions must be observed for these products:

The maintenance schedule is an integral part of the emissions certification. Nonobservance of the maintenance instructions can result in violation of the statutory emission specifications. Modification or removal of mechanical or electronic components or the installation of additional components as well as the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Emission-relevant components may only be serviced, replaced or repaired if components approved by MTU are used. Emission-relevant components are for instance control units, data records, sensors, injectors, exhaust flaps ands all components of exhaust aftertreatment systems.

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability.The maintenance intervals and the scope of required maintenance activities are the result of operational experience and therefore represent recommendations. Severe operating and ambient conditions may require additional maintenance work and/or modification of the maintenance intervals.

Adherence to the specified maintenance intervals is essential to maintain product safety.

Maintenance activities are specified with intervals based on hours in operation and time limits. Whichever value is reached first shall apply.

The individual maintenance intervals are classified in accordance with the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.
QL2: Exchange of components and parts (corrective only, not part of the maintenance schedule).
QL3: Maintenance work which requires partial disassembly of the product.
QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

For change intervals for fluids and lubricants as well as for preservation specifications, refer to the relevant instructions of the component manufacturers and the MTU Fluids and Lubricants Specifications. The latest version for drive systems is available online at: http://www.mtu-online.com, and for power generation systems at: http://www.mtuonsiteenergy.com.

Components which are not listed in this maintenance schedule must be serviced in accordance with Manufacturer's Instructions.

TIM-ID: 0000046248 - 006

1.2 Allocation matrix application group 4D

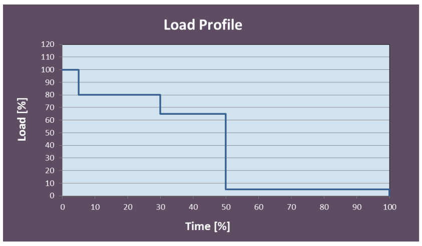
V 4000 S83 / S83 L								
Load profile		Load factor %	Load indicator %	0 - 140 kW/cyl.	141 - 156 kW/cyl.			
Load %	Time %							
110 (100-110)	–	0 - 41	≤ 9	13,500	10,000	–	–	
100 (90-100)	5							
90 (80-90)	–							
80 (65-80)	25							
65 (50-65)	20							
50 (30-50)	–							
30 (5-30)	–							
Idle (0-5)	50							

Table 2: Allocation matrix application group 4D

TIM-ID: 0000061001 - 001

1.3 Maintenance schedule matrix 13,500 h

0 to 13,500 Operating hours

Task	Limit	Operating hours [h]																											
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500
Engine																													
W0800	-																												
W1024	1 w																												
W1008	2 a																												
W0500	-	X																											
W0501	-	X																											
W0503	-	X																											
W0505	-	X																											
W0506	-	X																											
W0525	-	X																											
W1009	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W4172	1 a			X		X		X		X		X		X		X		X		X		X		X		X		X	
W1001	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X	
W1675	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X	
W1241	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X	
W1575	1 a							X						X						X						X			
W1207	2 a			X				X						X						X						X			
W1519	1 a										X									X									X
W1046	2 a										X									X									X
W1006	18 a										X									X									X
W1713	18 a										X									X									X
W1011	18 a										X									X									X
W1689	18 a										X									X									X
W4184	18 a										X									X									X
W1610	18 a																												X
W1250	5 a										X									X									X
W4150	5 a										X									X									X
W4170	5 a										X									X									X
W1251	6 a																												X
W2012	18 a																												X
W2001	18 a																												X
W2002	18 a																												X
W2004	18 a																												X
W2006	18 a																												X
W2007	18 a																												X
W2009	18 a																												X
W2010	18 a																												X
W2062	18 a																												X
W2070	18 a																												X
W2110	18 a																												X
W1041	18 a																												X
W1051	18 a																												X
W1043	18 a																												X

w = weeks
m = months
a = years

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Task		Operating hours [h]																													
	Limit	Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500		
W1735	18 a																												X		
W1134	18 a																												X		
W3000	18 a																												X		
W3001	18 a																												X		
W3002	18 a																												X		
W3003	18 a																												X		
W3004	18 a																												X		
W3005	18 a																												X		
W3007	18 a																												X		
W3018	18 a																												X		
W3021	18 a																												X		
W3083	18 a																												X		
W3103	18 a																												X		
W3108	18 a																												X		
W3143	18 a																												X		
W3177	18 a																												X		
W3182	18 a																												X		
W4173	18 a																												X		
W1124	18 a																												X		
W1331	18 a																												X		
Transmission (ZF)																															
X4032	-	X																													
X4033	-	X																													
X4034	-	X																													
X4035	-		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		
X4023	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		
X4037	-			X		X		X		X		X		X		X		X		X		X		X		X		X			
X4042	-																												X		
W4146	-																												X		
Air compressor																															
W1784	-										X									X									X		

w = weeks
m = months
a = years

TIM-ID: 0000064387 - 002

1.4 Maintenance tasks 13,500 h

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
Engine						
QL1	-	-	EPA label	Check if emission label is fitted and legible, and if it contains the correct information.		W0800
QL1	-	1 w	Emergency air shut-off flaps	Check operation of emergency air shut-off flaps (electrical).	X	W1024
QL1	-	2 a	Engine oil filter	Fit new engine oil filters each time the engine oil is changed or, at the latest, on expiry of the time limit (given in years).		W1008
QL1	Daily	-	Engine operation	Check engine oil level.		W0500
				Carry out visual inspection of engine for general condition and leaks.		W0501
				Inspect service indicator of air filter.		W0503
				Check relief bores of coolant pump(s).		W0505
				Check for abnormal running noises, exhaust gas color and vibration.		W0506
				Inspect battery-charging generator for contamination and clean if necessary.		W0525
QL1	500	2 a	Centrifugal oil filter	Check thickness of oil residue layer. Clean. Fit new sleeve, at the latest, each time the engine oil is changed.		W1009
QL1	1000	1 a	Engine mounts	Check general condition of resilient mount (visual inspection).		W4172
QL1	1000	2 a	Fuel filter	Fit new fuel filter or new fuel filter insert.		W1001
QL1	1000	2 a	Fuel prefilter	Fit new fuel prefilter or new fuel prefilter insert.		W1675
QL1	1000	2 a	Belt drive	Inspect condition of drive belts and fit new ones if necessary. Adjust tension.		W1241
QL1	3000	1 a	Engine mounts	Carry out visual inspection of Trunnion mounts for general condition.		W1575
QL1	3000	2 a	Valve gear	Check valve clearance, adjust if required. ATTENTION! Initial adjustment after 1,000 operating hours and subsequently 1,000 operating hours after each cylinder-head overhaul..		W1207
QL1	4500	1 a	Engine mounts	Check tightness of securing screws.		W1519
QL1	4500	2 a	Crankcase breathers	Fit new filters or filter inserts.		W1046
QL1	4500	18 a	Fuel injectors	Replace fuel injectors.		W1006
QL1	4500	18 a	Engine governor	Injector: reset drift compensation parameters (CDC).		W1713
QL1	4500	18 a	Combustion chambers	Inspect cylinder chambers using endoscope.		W1011
QL1	4500	18 a	Cylinder heads	Measure valve stem end to cylinder head top distance.		W1689
QL1	4500	18 a	Cradle securing screws	Check for secure seating.		W4184
QL1	13500	18 a	Engine mounts	Check securing screws of trunnion mounts for secure seating.		W1610
QL3	4500	5 a	Rubber sleeves	Replace all rubber sleeves.		W1250
QL3	4500	5 a	Engine mounts	Measure height of rubber element.	X	W4150
				Measure buffer clearance, adjust if needed.	X	W4170
QL3	13500	6 a	Hose lines	Replace all hose lines.		W1251
QL3	13500	18 a	Component maintenance	Before starting maintenance work, carry out test run and record operating parameters. Then drain coolant and flush cooling systems.		W2012
				Inspect rocker arms and valve bridge for wear. Insert an endoscope through the pushrod bore to visually inspect swing followers and camshaft running surfaces.		W2001
				Clean air ducting.		W2002
				Fit new high-pressure fuel sensor.		W2004
				Check engine coolant thermostat and fit new thermal actuator.		W2006
				Check charge-air coolant thermostat and fit new thermal actuator.		W2007
				Inspect centrifugal oil filter for wear.		W2009
				Overhaul starter.		W2010
				Fit new seals/sealing materials for all disassembled components.		W2062
w = weeks m = months a = years						

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Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
				Overhaul charge-air coolant pump.		W2070
				Overhaul engine coolant pump.		W2110
QL3	13500	18 a	Turbochargers	Fit new turbochargers.		W1041
QL3	13500	18 a	Fuel delivery pump	Fit new fuel delivery pump.		W1051
QL3	13500	18 a	Battery charging alternator	Overhaul battery-charging generator.		W1043
QL3	13500	18 a	Auxiliary PTO	Replace PTO.		W1735
QL3	13500	18 a	Cylinder heads	Overhaul cylinder heads.		W1134
QL4	13500	18 a	Extended component maintenance	Completely disassemble the engine.Inspect engine components as per assembly instructions and repair or fit new components as required.		W3000
				Replace all elastomeric parts and seals with new ones.		W3001
				Fit new piston rings.		W3002
				Fit new conrod bearings.		W3003
				Fit new crankshaft bearings.		W3004
				Fit new cylinder liners.		W3005
				Fit new high-pressure fuel pump.		W3007
				Fit new camshaft bearings and camshaft thrust bearings.		W3018
				Fit new pressure relief valve in high-pressure fuel system.		W3021
				Replace wiring harnesses.		W3083
				Replace swing followers and swing-follower shafts.		W3103
				Check vibration damper, fit new one if necessary.		W3108
				Replace pulley.		W3143
				Check engine oil pump, replace if necessary.		W3177
				Gear train: Check tooth flanks for wear (visual inspection), replace bearing bushings.		W3182
QL4	13500	18 a	Engine mounts	Replace rubber element of resilient mounts.		W4173
QL4	13500	18 a	Engine mounts	Fit new Trunnion mount element.		W1124
QL4	13500	18 a	Exhaust pipe bellows	Replace exhaust pipe bellows.		W1331
Transmission (ZF)						
QL1	Daily	-	Transmission operation	Check gear oil level.		X4032
				Check general condition of gearbox and inspect for leaks.		X4033
				Check for abnormal running noise and vibration.		X4034
QL1	500	-	Gear oil filter	Replace transmission oil filter.		X4035
QL1	500	1 a	Torsional resilient coupling	Check torsionally resilient coupling (visual inspection).		X4023
QL1	1000	-	Connectors/wiring harness	Check all electrical wiring components for chafing or damage.		X4037
QL4	13500	-	Transmission	Overhaul gearbox.		X4042
QL4	13500	-	Flexible/torsional resilient coupling	Replace rubber element.		W4146
Air compressor						
QL3	4500	-	Air compressor	Check air compressor and replace if required.	X	W1784
w = weeks m = months a = years						

TIM-ID: 0000064388 - 002

1.5 Maintenance schedule matrix 10,000 h

0 to 10,000 Operating hours

Task		Operating hours [h]																				
	Limit	Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000
Engine																						
W0800	-																					
W1024	1 w																					
W1008	2 a																					
W0500	-	X																				
W0501	-	X																				
W0503	-	X																				
W0505	-	X																				
W0506	-	X																				
W0525	-	X																				
W1009	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W4172	1 a			X		X		X		X		X		X		X		X		X		X
W1001	2 a			X		X		X		X		X		X		X		X		X		X
W1675	2 a			X		X		X		X		X		X		X		X		X		X
W1241	2 a			X		X		X		X		X		X		X		X		X		X
W1575	1 a						X					X					X					X
W1207	2 a			X			X					X					X					X
W1519	1 a											X										X
W1046	2 a											X										X
W1006	18 a											X										X
W1713	18 a											X										X
W1011	18 a											X										X
W1689	18 a											X										X
W4184	18 a											X										X
W1610	18 a																					X
W1250	5 a											X										X
W4150	5 a											X										X
W4170	5 a											X										X
W1041	18 a											X										X
W1251	6 a																					X
W2012	18 a																					X
W2001	18 a																					X
W2002	18 a																					X
W2004	18 a																					X
W2006	18 a																					X
W2007	18 a																					X
W2009	18 a																					X
W2010	18 a																					X
W2062	18 a																					X
W2070	18 a																					X
W2110	18 a																					X
W1051	18 a																					X
W1043	18 a																					X

w = weeks
m = months
a = years

TIM-ID: 0000064389 - 002